



Name: _____

Teacher: _____

Knox Grammar School

2015

**Trial Higher School Certificate
Examination**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen only
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Subject Teachers

Mr M Vuletich
Mr I Bradford

Setter

Mr M Vuletich

**This paper MUST NOT be removed from the
examination room**

Number of Students in Course: 35

Total Marks – 100

Section I 10 Marks

- Answer Questions 1 to 10
- Use the Multiple Choice Answer Sheet

Section II 90 Marks

- Answer Questions 11 to 16
- All questions are worth 15 marks
- Answer each question in a separate Writing Booklet.

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Section I

10 Marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1 Which of the following is an expression for $\int \frac{1}{1 + \sin x + \cos x} dx$?

- (A) $\ln|t-1| + C$
- (B) $\ln|t+1| + C$
- (C) $\ln|t^2-1| + C$
- (D) $\ln|t^2+1| + C$

2 What is the eccentricity for the hyperbola $\frac{y^2}{225} - \frac{x^2}{64} = 1$?

- (A) $\frac{8}{17}$
- (B) $\frac{15}{17}$
- (C) $\frac{17}{15}$
- (D) $\frac{17}{8}$

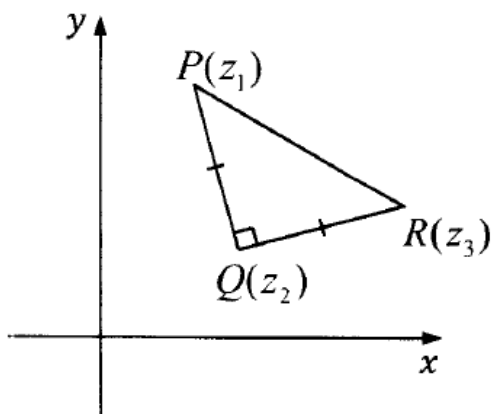
3 The polynomial equation $x^3 - 5x^2 + 6 = 0$ has roots α , β and γ . Which of the following polynomial equations has roots α^2 , β^2 and γ^2 ?

- (A) $x^3 - 25x^2 + 60x - 36 = 0$
- (B) $x^3 - 25x^2 + 60x - 12 = 0$
- (C) $x^3 - x^2 + 12x - 36 = 0$
- (D) $x^3 - x^2 + 12x - 12 = 0$

- 4 Given that $z - 2 + i$ is a factor of $P(z) = 2z^3 - 7z^2 + 6z + 5$ over the complex field, which one of the following statements must be true?

- (A) $P(-2 + i) = 0$
- (B) $P(-2 - i) = 0$
- (C) The equation $P(z) = 0$ has one complex and two real roots
- (D) The equation $P(z) = 0$ has one real and two complex roots

- 5 The vertices of $\triangle PQR$ are represented by the complex numbers z_1, z_2 and z_3 respectively. The $\triangle PQR$ is isosceles and right-angled at Q , as shown in the diagram.



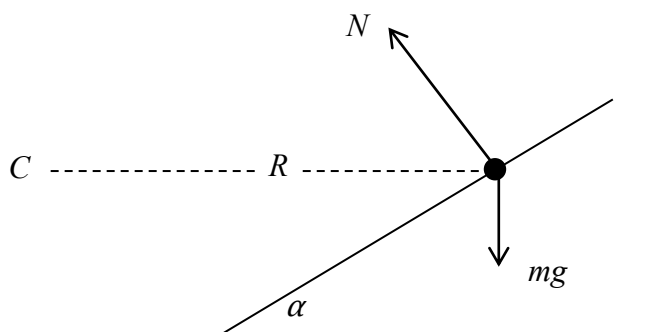
Which of the following statements is true?

- (A) $z_2 - z_1 = i(z_3 - z_2)$
- (B) $z_1 - z_2 = i(z_3 - z_2)$
- (C) $z_2 - z_1 = i(z_1 - z_3)$
- (D) $z_1 - z_2 = i(z_1 - z_3)$

- 6 Using the binomial theorem $(1+x)^n = {}^nC_0 + {}^nC_1x^1 + {}^nC_2x^2 + \dots + {}^nC_nx^n = \sum_{k=0}^n {}^nC_kx^k$, which of the following expressions is correct?

- (A) $(1 + \frac{1}{n})^n = \sum_{k=0}^n \frac{n(n-1)(n-2)\dots(n-k)}{n^k} \times \frac{1}{k!}$
- (B) $(1 + \frac{1}{n})^n = \sum_{k=0}^n \frac{n(n-1)(n-2)\dots(n-k)}{n^k} \times \frac{1}{(k+1)!}$
- (C) $(1 + \frac{1}{n})^n = \sum_{k=0}^n \frac{n(n-1)(n-2)\dots(n-k+1)}{n^k} \times \frac{1}{k!}$
- (D) $(1 + \frac{1}{n})^n = \sum_{k=0}^n \frac{n(n-1)(n-2)\dots(n-k+1)}{n^k} \times \frac{1}{(k+1)!}$

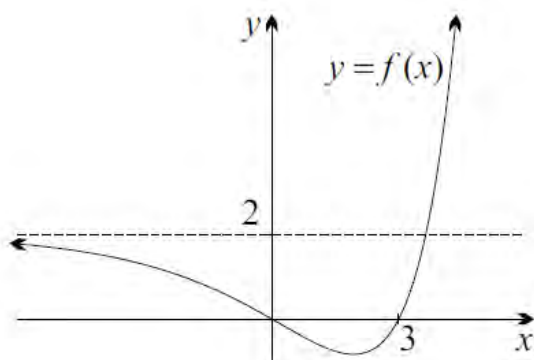
- 7 A particle of mass, m , travels with constant velocity, v , in a horizontal circle of radius, R , centre, C , around a track banked at an angle, α , to the horizontal, as shown in the diagram. There is no tendency for the particle to slip sideways.



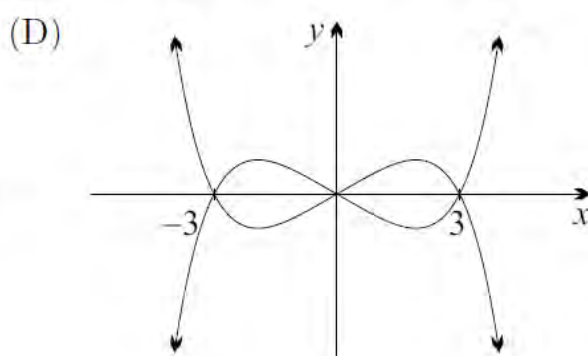
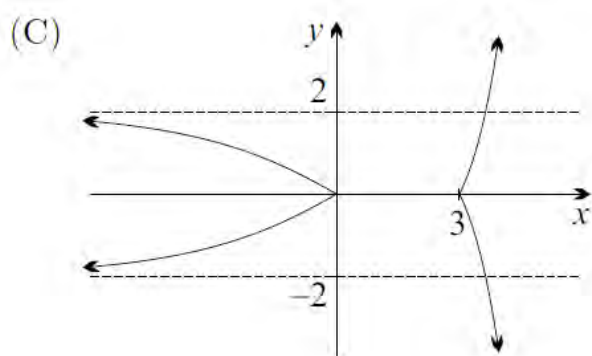
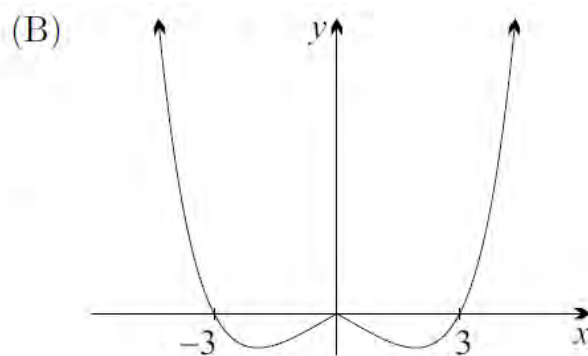
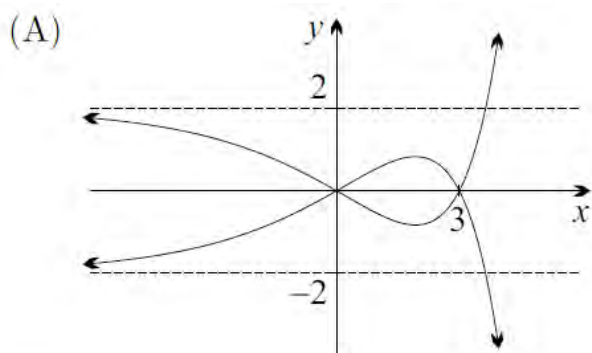
What is the expression for the vertical component of the forces acting on the particle?

- (A) $N \cos \alpha = mg$
- (B) $N \sin \alpha = \frac{mv^2}{R}$
- (C) $N \sin \alpha = mg$
- (D) $N \cos \alpha = \frac{mv^2}{R}$

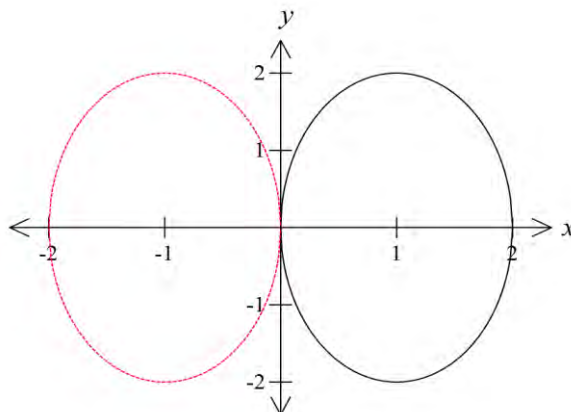
- 8 The graph of $y = f(x)$ is shown below.



Which is the correct graph of $|y| = f(x)$?



- 9 The region enclosed by the ellipse $(x-1)^2 + \frac{y^2}{4} = 1$ is rotated about the y axis to form a solid.



If slices are taken perpendicular to the axis of rotation, what is the correct expression for the volume?

- (A) $V = \int_{-2}^2 \pi \sqrt{1-y^2} dy$
- (B) $V = \int_{-2}^2 2\pi \sqrt{1-y^2} dy$
- (C) $V = \int_{-2}^2 \pi \sqrt{4-y^2} dy$
- (D) $V = \int_{-2}^2 2\pi \sqrt{4-y^2} dy$
- 10 How many six letter words can be formed using the letters of the word 'PRESSES'?
- (A) 420
- (B) 120
- (C) 300
- (D) 240

End of Section I

Section II

90 Marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

All necessary working should be shown in every question.

Question 11 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) Let $z = 2 + 2i$ and $\omega = 1 + 3i$.

(i) Write $z - \omega$ in modulus-argument form. **2**

(ii) Find $\overline{\left(\frac{z}{\omega}\right)}$ in the form $a + bi$ where a and b real. **2**

(b) Evaluate $\int_1^e x^7 \ln x \, dx$. **3**

(c) Sketch the region in the complex plane where the inequalities

$|z - 1 - i| \leq \sqrt{2}$ and $0 \leq \arg(z - 1 - i) \leq \frac{\pi}{4}$ are satisfied simultaneously. **3**

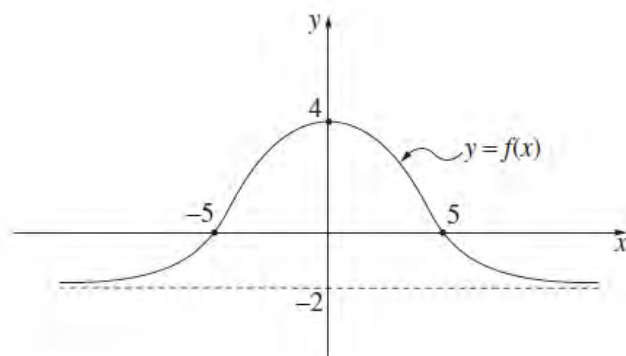
(d) Without the use of calculus, sketch the graph of $y = \frac{x^3 + 1}{x}$, showing any asymptotes and intercepts with the coordinate axes. **2**

(e) The area defined by $0 \leq y \leq \sin x$ for $0 \leq x \leq \pi$ is rotated about the y -axis. Use the method of cylindrical shells to find the volume of the resulting solid of revolution. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) The diagram shows the graph of $y = f(x)$.

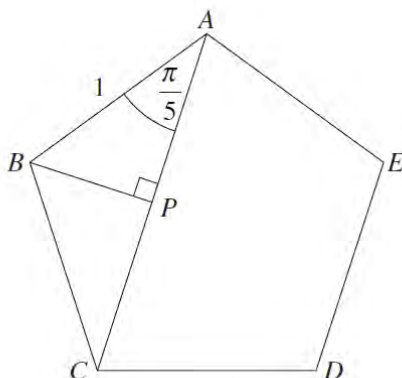


Draw a separate half-page graph for each of the following functions, showing all asymptotes and intercepts.

(i) $y = \frac{1}{\sqrt{f(x)}}$ 2

(ii) $y = xf(x)$ 2

- (b)



In the diagram, $ABCDE$ is a regular pentagon with sides of length 1 unit. The perpendicular to AC through B meets AC at P .

Copy or trace the diagram into your writing booklet.

(i) Let $u = \cos \frac{\pi}{5}$. Use the cosine rule in $\triangle ACD$ to show that $8u^3 - 8u^2 + 1 = 0$. 2

(ii) One root of $8x^3 - 8x^2 + 1 = 0$ is $\frac{1}{2}$. By finding the other roots of

$8x^3 - 8x^2 + 1 = 0$, find the exact value of $\cos \frac{\pi}{5}$. 2

Question 12 continues on page 9

Question 12 (continued)

(c) Find the equation of the tangent to the curve $x^2 - xy + y^3 = 5$ at the point $(2, -1)$. **3**

(d) Let $I_n = \int_0^1 \frac{x^{2n}}{x^2 + 1} dx$ for integers $n \geq 0$.

(i) Show that $I_n + I_{n-1} = \frac{1}{2n-1}$ for integers $n \geq 1$. **2**

(ii) Hence, or otherwise, find $\int_0^1 \frac{x^4}{x^2 + 1} dx$. **2**

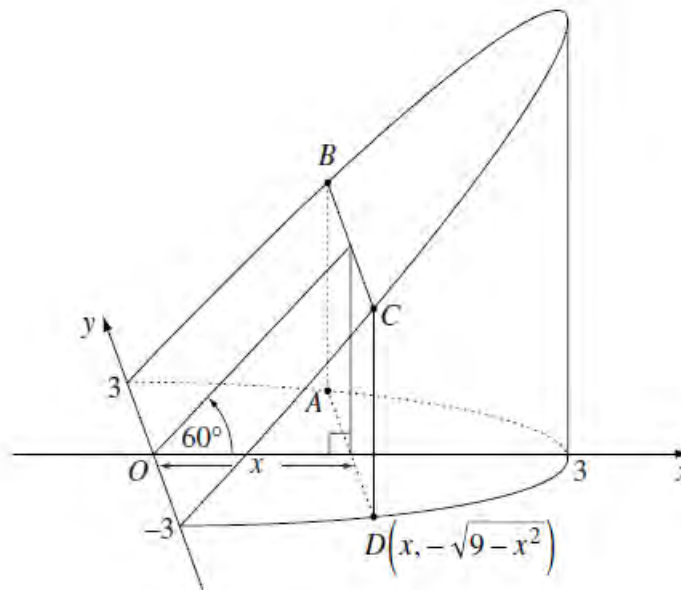
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Find $\int \frac{5x^2 - 3x + 13}{(x-1)(x^2 + 4)} dx$.

3

- (b) The base of a right cylinder is a circle in the xy plane with centre O and radius 3. A wedge is obtained by cutting this cylinder with the sloping plane through the y -axis inclined at 60° to the xy plane, as shown in the diagram.



A rectangular slice $ABCD$ is taken perpendicular to the base of the wedge at a distance x from the y -axis.

(i) Show that the area of $ABCD$ is given by $2x\sqrt{27-3x^2}$.

2

(ii) Find the volume of the wedge.

2

Question 13 continues on page 11

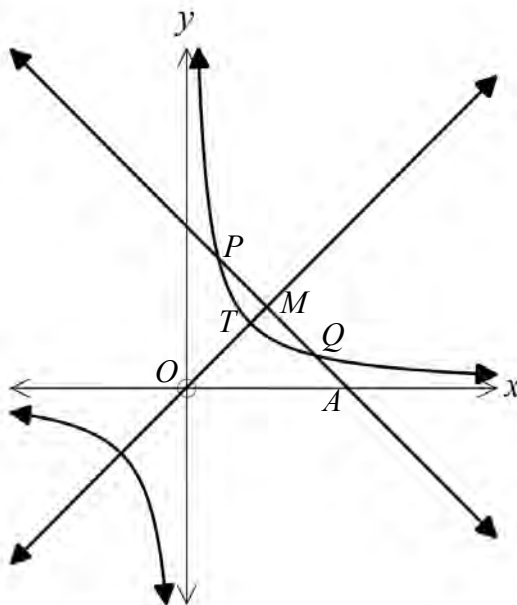
Question 13 (continued)

- (c) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $0 < b < a$, cuts the x -axis at A and A' .
The ellipse has eccentricity e and $S(ae, 0)$ is the focus of ellipse nearer to A .
The focal chord PSQ is perpendicular to the x -axis.

(i) Draw a diagram to represent this information. 1

(ii) Show that $\frac{1}{AS} + \frac{1}{A'S} = \frac{4}{PQ}$. 2

- (d) In the diagram $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ where $0 < p < q$,
are points on the hyperbola $xy = c^2$.
 M is the midpoint of PQ and the line PQ cuts the x -axis at A .
 OM cuts the hyperbola at T .



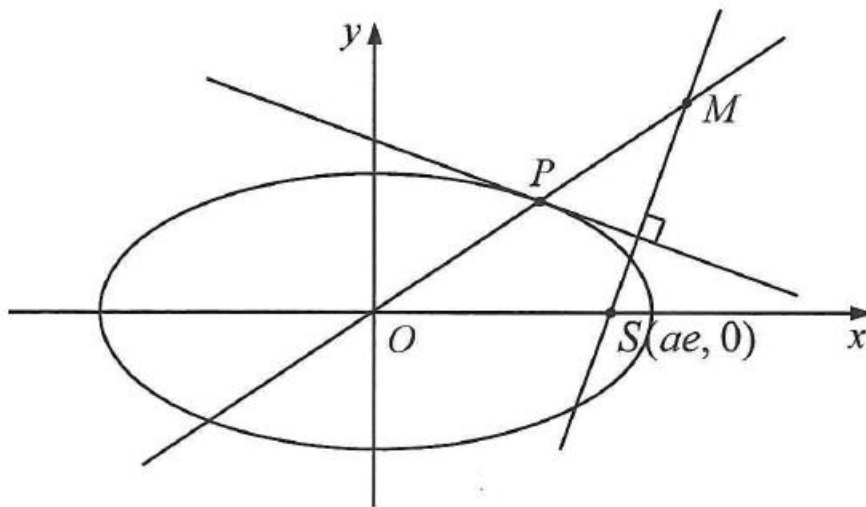
- (i) Show that gradient $OM = -1 \times$ gradient MA . 2
- (ii) Hence show that $OM = MA$. 1
- (iii) Show that the tangent to the hyperbola at T is parallel to the chord PQ . 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) The polynomial $P(x) = px^3 - 3qx + r$ has a zero multiplicity 2. **3**
Show that $4q^3 = pr^2$.

- (b) The diagram shows the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with focus $S(ae, 0)$ and centre O .
 $P(a \cos \theta, b \sin \theta)$ is any point on the ellipse. The line through S perpendicular to the tangent at P and the line OP produced meet at M .



- (i) Show the gradient of the tangent at P is given by $-\frac{b \cos \theta}{a \sin \theta}$. **1**
- (ii) Show that M lies on the corresponding directrix to the focus at S . **3**
- (c) A group of 30 students is to be divided into three groups consisting of 7, 8 and 15 students. In how many ways can this be done? Leave your answer in unsimplified form. **1**

Question 14 continues on page 13

Question 14 (continued)

- (d) An object of mass 2 kg is projected vertically upwards from ground level at a speed of 20 m/s. It experiences a resistance of $\frac{v^2}{2}$ Newtons at a speed of v m/s, and reaches a maximum height of H metres. Take upwards as positive and $g = 10\text{m/s}^2$.
- (i) If x is the displacement of the object from ground level after t seconds, show that its acceleration is given by $\ddot{x} = \frac{-40 - v^2}{4}$. 1
- (ii) Show that it takes approximately 0.8 seconds for the object to reach its maximum height. 2
- (iii) Find the maximum height H reached by the object. 2
- (iv) Calculate the speed of projection required to reach a maximum height of $2H$ metres. 2

End of Question 14

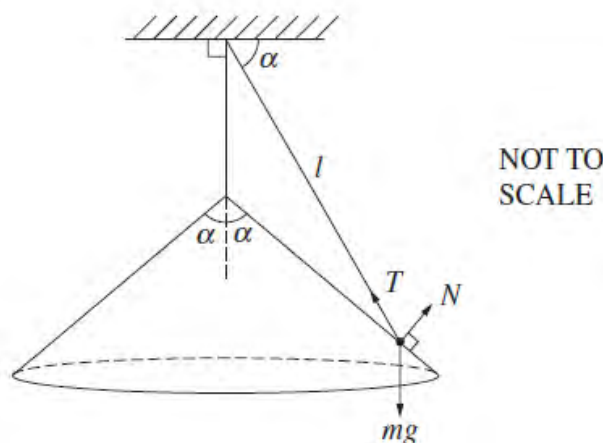
Question 15 (15 marks) Use a SEPARATE writing booklet

- (a) (i) Use calculus to show that $x > \ln(1+x)$ for all $x > 0$. 2

- (ii) Use the inequality in part (i) and the principle of mathematical induction to prove that

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} > \ln(1+n) \text{ for all positive integers } n. \quad 3$$

- (b) A particle of mass m is suspended by a string of length l from a point directly above the vertex of a smooth cone, which has a vertical axis. The particle remains in contact with the cone and rotates as a conical pendulum with angular velocity ω . The angle of the cone at its vertex is 2α , where $\alpha > \frac{\pi}{4}$, and the string makes an angle of α with the horizontal as shown in the diagram. The forces acting on the particle are the tension in the string T , the normal reaction to the cone N and the gravitational force mg .



- (i) Show, with the aid of a diagram, that the vertical component of N is $N \sin \alpha$. 1
- (ii) Show that $T + N = \frac{mg}{\sin \alpha}$, and find an expression for $T - N$ in terms of m , l and ω . 3
- (iii) The angular velocity is increased until $N = 0$, that is, when the particle is about to lose contact with the cone. Find an expression for this value of ω in terms of α , l and g . 2

Question 15 continues on page 15

Question 15 (continued)

- (c) (i) It can be shown that for positive real numbers a and b that

$$a^2 + b^2 \geq 2ab \quad (\text{DO NOT PROVE THIS}).$$

Hence show for positive real numbers a, b, c and d that:

$$3(a^2 + b^2 + c^2 + d^2) \geq 2(ab + ac + ad + bc + bd + cd) \quad \mathbf{2}$$

- (ii) Hence show for positive real numbers a, b, c and d that

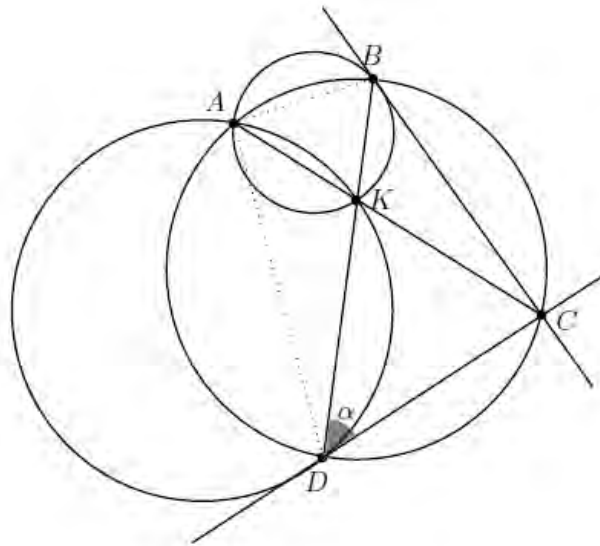
$$\text{if } a + b + c + d = 1 \text{ then } ab + ac + ad + bc + bd + cd \leq \frac{3}{8}. \quad \mathbf{2}$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) There are 40 balls in a game of lotto numbered from 1 to 40. Six balls are selected at random.
- (i) What is the probability, correct to four decimal places, of drawing the number 7 in exactly two of the next five games? 2
 - (ii) What is the probability, correct to four decimal places, of drawing the number 7 in at least two of the five games? 1
 - (iii) What is the probability that the number 7 is drawn and it is the highest number drawn in at least one of the next five games? 2
Give your answer in scientific notation correct to three significant figures.

(b)



In the diagram above, $ABCD$ is a cyclic quadrilateral and diagonals AC and BD intersect at K . Circles AKD and AKB are drawn and it is known that CD is a tangent to circle AKD . Let $\angle CDB = \alpha$.

Use the separate blue answer sheet for Question 16 (b).

- (i) Prove that $\triangle BCD$ is isosceles. 2
- (ii) Prove that CB is a tangent to circle AKB 2

Question 16 is continued on page 17

Question 16 (continued)

(c) Suppose that x is a positive real number.

(i) Show that $\frac{1}{1+t^2} < 1 - t^2 + t^4 - t^6 + \dots + t^{4n} < \frac{1}{1+t^2} + t^{4n+2}$, for $0 < t < x$. **3**

(ii) Hence show that:

$$\tan^{-1} x < x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + \frac{x^{4n+1}}{4n+1} < \tan^{-1} x + \frac{x^{4n+3}}{4n+3} \quad \mathbf{1}$$

(iii) Explain why $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ for $0 \leq x \leq 1$. **1**

(iv) Deduce that $\pi = 4 \times (1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots)$. **1**

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2015 MATHEMATICS EXTENSION 2
TRIAL HSC SOLUTIONS

①

1. B 2. C 3. A 4. D 5. B
6. C 7. A 8. C 9. D 10. A

Question 1

$$\begin{aligned}\int \frac{1}{1 + \sin x + \cos x} dx &= \int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\ &= \int \frac{2}{1+t^2+2t+1-t^2} dt \\ &= \int \frac{1}{1+t} dt \\ &= \ln|t+1| + C \quad \therefore B\end{aligned}$$

Question 2

$$\begin{aligned}\text{For } \frac{y^2}{225} - \frac{x^2}{64} &= 1 \quad a=9, b=15 \\ a^2 &= b^2(e^2-1) \\ \therefore e^2 &= 1 + \frac{64}{225} \\ e &= \frac{17}{15} \quad \therefore C\end{aligned}$$

Question 3

let $x = \sqrt{y}$ for roots $\alpha^2, \beta^2, \gamma^2$

$$\begin{aligned}\therefore (\sqrt{y})^3 - 5(\sqrt{y})^2 + 6 &= 0 \\ y^{\frac{3}{2}} &= 5y - 6 \\ \therefore y^3 &= 25y^2 - 60y + 36 \\ \therefore y^3 - 25y^2 + 60y - 36 &= 0 \quad \therefore A\end{aligned}$$

②

Question 4

$P(z)$ has real co-efficients $\therefore$ complex roots occur in conjugate pairs. Since $P(z)$ is of degree 3, it must have 2 complex and one real root $\therefore D$

Question 5

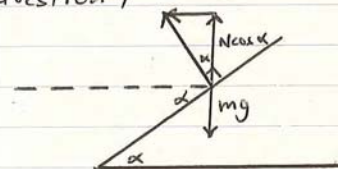
$\vec{QP} = i \times \vec{QR}$ as $|QP| = |QR|$ and $\angle PQR = 90^\circ$
and QR is rotated anti-clockwise about Q

$$\therefore z_1 - z_2 = i(z_3 - z_2) \quad \therefore B$$

Question 6

$$\begin{aligned}\left(1 + \frac{1}{n}\right)^n &= \sum_{k=0}^n {}^nC_k \left(\frac{1}{n}\right)^k \\ &= \sum_{k=0}^n \frac{n!}{k!(n-k)!} \cdot \frac{1}{n^k} \\ &= \sum_{k=0}^n \frac{n(n-1)(n-2)\dots 3 \times 2 \times 1}{n^k (n-k)(n-k-1)\dots 3 \times 2 \times 1} \cdot \frac{1}{k!} \\ &= \sum_{k=0}^n \frac{n(n-1)(n-2)\dots (n-k+1)}{n^k} \cdot \frac{1}{k!} \quad \therefore C\end{aligned}$$

Question 7



Resolving vertical forces
 $N \cos \alpha = mg \quad \therefore A$

Question 8

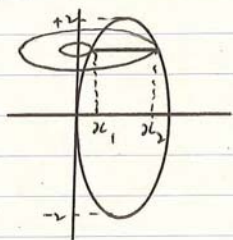
If $|y| = f(x)$ then $f(x) \geq 0$

Apply definition $|y| = \sqrt{y^2}$

$$\sqrt{y^2} = f(x)$$

$y = \pm f(x)$ only for $f(x) \geq 0 \quad \therefore C$

Question 9



$$\Delta V = \pi(R+r)(R-r) \cdot \Delta y$$

where $R = x_2, r = x_1$

Taking x_2, x_1 as roots of equation $(x-1)^2 + y^2 = 1$ at some fixed height y above x -axis:

$$4(x^2 - 2x + 1) + y^2 - 4 = 0$$

$$4x^2 - 8x + y^2 = 0 \text{ has roots } x_1, x_2$$

$$x_1 + x_2 = 2$$

$$x_2 - x_1 = \sqrt{(x_1 + x_2)^2 - 4x_1x_2} \quad x_2 - x_1 > 0$$

$$= \sqrt{4 - 4 \cdot \frac{y^2}{4}}$$

$$= \sqrt{4 - y^2}$$

$$\therefore V = 2\pi \int_{-2}^2 \sqrt{4 - y^2} dy \quad \therefore D$$

Question 10

Using cases:

Omit S : $\frac{6!}{2! \times 2!} = 180 \quad (2Es, 2Ss)$

Omit E : $\frac{6!}{3!} = 120 \quad (3Ss)$

$$\therefore \text{Total} = 420$$

Omit P : $\frac{6!}{3! \times 2!} = 60 \quad (2Es, 3Ss)$

$\therefore A$

Omit R : $\frac{6!}{3! \times 2!} = 60 \quad (2Es, 3Ss)$

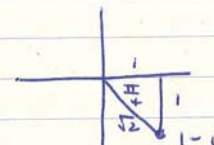
Section II

Question 11

(a) (i) $z - w = 2 + 2i - (1 + 3i)$

$$= 1 - i$$

$$= \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$$



(ii) $\overline{\left(\frac{z}{w}\right)} = \overline{\left(\frac{2+2i}{1+3i} \times \frac{1-3i}{1-3i}\right)}$

$$= \overline{\left(\frac{2 - 4i + 6}{10}\right)}$$

$$= \frac{4}{5} - \frac{2}{5}i$$

$$= \frac{4}{5} + \frac{2}{5}i$$

(b) $I = \int_1^e x^7 \ln x dx \quad \left. \begin{array}{l} u = \ln x \quad v' = x^7 \\ u' = \frac{1}{x} \quad v = \frac{x^8}{8} \end{array} \right\} \checkmark$

$$= \left[\frac{x^8 \ln x}{8} \right]_1^e - \frac{1}{8} \int_1^e x^7 dx \quad \checkmark$$

$$= \frac{e^8}{8} - \frac{1}{8} \left[\frac{x^8}{8} \right]_1^e$$

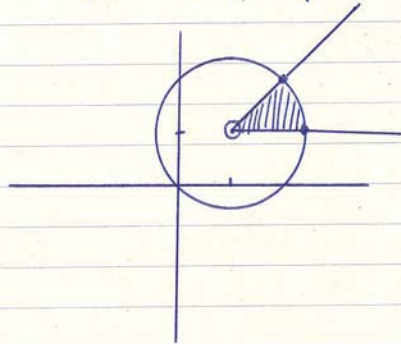
$$= \frac{e^8}{8} - \frac{1}{8} \left(\frac{e^8}{8} - \frac{1}{8} \right)$$

$$= \frac{7e^8 + 1}{64} \quad \checkmark \text{ or equivalent}$$

5

(c) $|z - (1+i)| \leq 2$

$0 \leq \arg(z - (1+i)) \leq \frac{\pi}{4}$



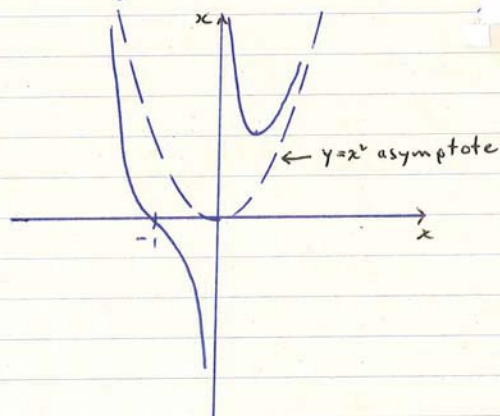
- ✓ circle
- ✓ sector
- ✓ shaded region

(d) $y = \frac{x^3 + 1}{x}$
 $= x^2 + \frac{1}{x}$

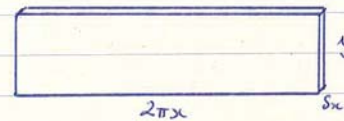
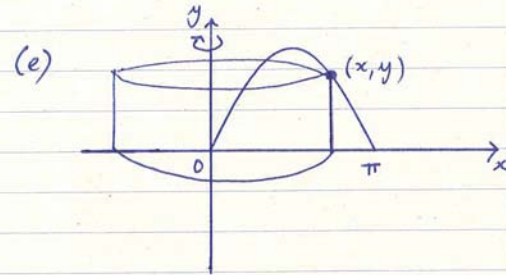
At $x=0$, y undefined
 $y=0$, $x=-1$

Vertical Asymptote at $x=0$ as $x \rightarrow 0^+$, $y \rightarrow \infty$
as $x \rightarrow 0^-$, $y \rightarrow -\infty$

as $x \rightarrow \infty$ $y \rightarrow (x^2)^+$
as $x \rightarrow -\infty$ $y \rightarrow (x^2)^-$



6



$\therefore \delta V = 2\pi x y \delta x$

$\therefore V = 2\pi \int_0^\pi x y dx$

$= 2\pi \int_0^\pi x \sin x dx$

$u = x \quad v' = \sin x$
 $u' = 1 \quad v = -\cos x$

$= 2\pi \left\{ [-x \cos x]_0^\pi + \int_0^\pi \cos x dx \right\}$

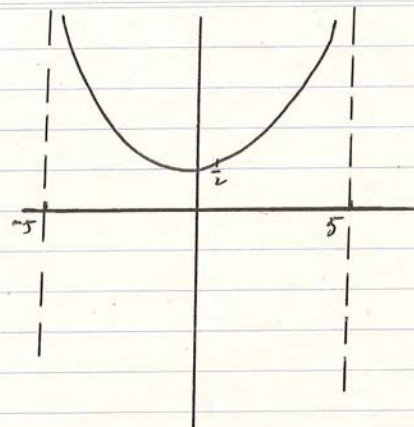
$= 2\pi \left\{ \pi + [\sin x]_0^\pi \right\}$

$= 2\pi^2 \text{ units}^3$

7

Question 12

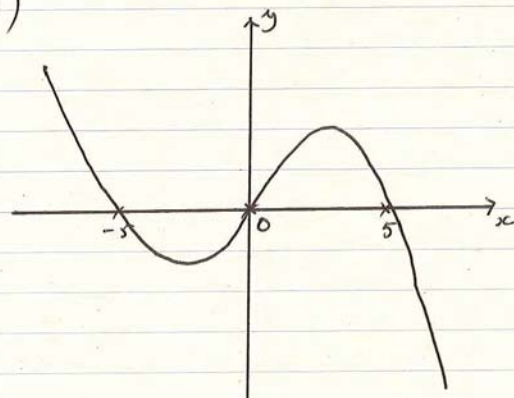
(a) i)



✓ asymptotes and intercept

✓ shape.

ii)



✓ intercepts

✓ shape.

8

$$(b) (i) \cos \frac{\pi}{5} = AP$$

$$\therefore AP = u$$

$$\triangle APB \equiv \triangle CPB \text{ (RHS)}$$

$$\therefore PC = u$$

$$\therefore AC = 2u$$

$$\angle PAE = \frac{\pi}{5} \text{ (corresponding angle of congruent } \triangle \text{ s } ABC \text{ and } AED \text{ (SAS))}$$

$$\therefore \angle CAD = \frac{3\pi}{5} - \frac{\pi}{5} - \frac{\pi}{5} \text{ (Interior } \angle \text{ of regular pentagon is } \frac{3\pi}{5})$$

$$= \frac{\pi}{5}$$

$$\therefore CD^2 = AC^2 + AD^2 - 2 \times AC \times AD \times \cos \frac{\pi}{5}$$

$$1 = (2u)^2 + (2u)^2 - 2(2u)(2u) \cos \frac{\pi}{5}$$

$$1 = 4u^2 + 4u^2 - 8u^2 \text{ as } \cos \frac{\pi}{5} = u$$

$$\therefore 8u^3 - 8u^2 + 1 = 0 \text{ as required}$$

✓ correct use of cosine rule.

progress to setup.

$$(ii) \text{ So } \frac{1}{2} + \alpha + \beta = 1 \text{ (sum of roots)}$$

$$\alpha + \beta = \frac{1}{2}$$

$$\frac{1}{2} \alpha \beta = -\frac{1}{8} \text{ (product of roots)}$$

$$\alpha \beta = -\frac{1}{4}$$

$$\therefore \alpha - \frac{1}{4\alpha} = \frac{1}{2}$$

$$4\alpha^2 - 1 = 2\alpha$$

$$4\alpha^2 - 2\alpha - 1 = 0$$

$$\alpha = \frac{2 \pm \sqrt{4 + 16}}{8}$$

$$= \frac{1 \pm \sqrt{5}}{4}$$

$$\therefore \cos \frac{\pi}{5} = \frac{1 + \sqrt{5}}{4} \text{ as } \cos \frac{\pi}{5} > 0$$

✓ significant progress

(9)

$$(c) \quad \frac{d}{dx} x^2 - \frac{d}{dx} xy + \frac{dy^3}{dx} = 5$$

$$2x - (x \cdot \frac{dy}{dx} + y) + 3y^2 \frac{dy}{dx} = 0$$

$$2x - x \frac{dy}{dx} - y + 3y^2 \frac{dy}{dx} = 0 \quad \checkmark$$

$$\frac{dy}{dx} (3y^2 - x) = y - 2x$$

$$\therefore \frac{dy}{dx} = \frac{y-2x}{3y^2-x} \quad \checkmark$$

$$\text{at } (2, -1) \quad \frac{dy}{dx} = \frac{-1-4}{3-2} = -5$$

$$\therefore y+1 = -5(x-2)$$

$$y+1 = -5x+10$$

$$5x+y-9=0 \text{ is tangent} \quad \checkmark$$

$$(d) i) \quad I_n + I_{n-1} = \int_0^1 \frac{x^{2n}}{x^2+1} dx + \int_0^1 \frac{x^{2(n-1)}}{x^2+1} dx$$

$$= \int_0^1 \frac{x^{2n} + x^{2n-2}}{x^2+1} dx$$

$$= \int_0^1 \frac{x^{2n} + \frac{x^{2n}}{x^2}}{x^2+1} dx$$

$$= \int_0^1 \frac{x^2 x^{2n} + x^{2n}}{x^2(x^2+1)} dx$$

$$= \int_0^1 \frac{x^{2n}}{x^2} dx$$

$$= \int_0^1 x^{2n-2} dx$$

$$= \left[\frac{x^{2n-1}}{2n-1} \right]_0^1 \quad \checkmark$$

$$= \frac{1}{2n-1}$$

significant
progress

(10)

$$(d) ii) \quad \int_0^1 \frac{x^4}{x^2+1} dx = I_2$$

$$= \frac{1}{2(2)-1} - I_1$$

$$= \frac{1}{3} - \left[\frac{1}{2(1)-1} - I_0 \right]$$

$$= \frac{1}{3} - 1 + I_0 \quad \text{and } I_0 = \int_0^1 \frac{1}{x^2+1} dx$$

$$= \frac{\pi}{4} - \frac{2}{3} \quad \checkmark$$

$$= \tan^{-1} x \Big|_0^1$$

$$= \frac{\pi}{4}$$

Question 13

$$(a) \quad \text{let } \frac{5x^2-3x+13}{(x-1)(x^2+4)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$$

$$\equiv \frac{A(x^2+4) + (Bx+C)(x-1)}{(x-1)(x^2+4)}$$

$$\therefore 5x^2-3x+13 = A(x^2+4) + (Bx+C)(x-1) \quad \checkmark$$

$$x=1 \quad \therefore 15 = 5A$$

$$A = 3$$

$$x=0 \quad \therefore 13 = 12 - C$$

$$\therefore C = -1$$

$$x=-1 \quad \therefore 21 = 15 + (-B-1) \cdot -2$$

$$6 = 2B+2$$

$$\therefore B = 2$$

$$\therefore I = \int \frac{3}{x-1} + \frac{2x-1}{x^2+4} dx$$

$$= 3 \ln|x-1| + \int \frac{2x}{x^2+4} dx - \int \frac{1}{x^2+4} dx$$

$$= 3 \ln|x-1| + \ln|x^2+4| - \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C$$

or equivalent

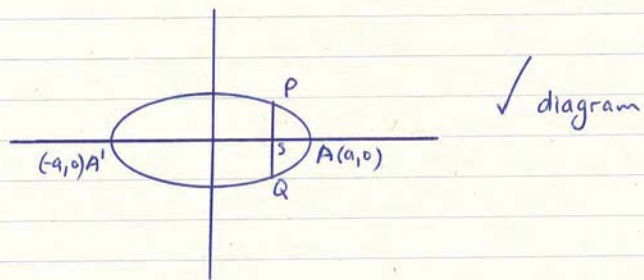
(11)

(b) (i) Area = AD × CD
 $= 2y \times CD$
 $= 2y \times \sqrt{3}x$ ✓ $\left. \begin{array}{l} \tan 60^\circ = \frac{CD}{x} \\ CD = \sqrt{3}x \end{array} \right\}$ ✓
 $= 2\sqrt{9-x^2} \cdot \sqrt{3}x$
 $= 2x\sqrt{3(9-x^2)}$
 $= 2x\sqrt{27-3x^2}$

ii) $\delta V = 2x\sqrt{27-3x^2} \cdot \delta x$

$\therefore V = \int_0^3 2x(27-3x^2)^{\frac{1}{2}} dx$
 $= -\frac{1}{3} \int_0^3 6x(27-3x^2)^{\frac{1}{2}} dx$ ✓
 $= -\frac{1}{3} \left[\frac{(27-3x^2)^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^3$
 $= -\frac{2}{9} \left[0 - 27^{\frac{3}{2}} \right]$
 $= \frac{2}{9} \times (\sqrt{27})^3$
 $= \frac{2}{9} \times (3\sqrt{3})^3$
 $= \frac{2}{9} \times 27 \times 3\sqrt{3}$
 $= 18\sqrt{3} \text{ units}^3$ ✓

(c) i)



(12)

ii) L.H.S = $\frac{1}{As} + \frac{1}{A's}$
 $= \frac{1}{a-ae} + \frac{1}{ae+a}$
 $= \frac{ae+a+a-ae}{a^2-a^2e^2}$
 $= \frac{2a}{a^2(1-e^2)} = \frac{2a}{b^2}$ ✓ $\left. \begin{array}{l} \text{For ellipse} \\ \frac{b^2}{a^2} = 1-e^2 \end{array} \right\}$

R.H.S = $\frac{4}{PQ}$ at $x=ae$ $\frac{a^2e^2}{a^2} + \frac{y^2}{b^2} = 1$
 $= \frac{4}{2b\sqrt{1-e^2}}$ $\therefore y^2 = b^2(1-e^2)$
 $= \frac{4}{2b} \cdot \frac{1}{\frac{b}{a}}$ $y = \pm b\sqrt{1-e^2}$
 $= \frac{4a}{2b^2}$ $\therefore Q(ae, -b\sqrt{1-e^2})$
 $= \frac{2a}{b^2}$ ✓ $P(ae, b\sqrt{1-e^2})$

$\therefore \frac{1}{As} + \frac{1}{A's} = \frac{4}{PQ}$

(d) i) $M\left(\frac{cp+cq}{2}, \frac{c}{\frac{p}{q}+\frac{q}{p}}\right)$

$= M\left(\frac{c}{2}(p+q), \frac{c}{2}\left(\frac{p+q}{pq}\right)\right)$

$\therefore m_{OM} = \frac{\frac{c}{2}(p+q)}{\frac{c}{2}(p+q)} = \frac{1}{pq}$ ✓

$m_{MA} = m_{PQ} = \frac{\frac{c}{q} - \frac{c}{p}}{cq - cp} = \frac{c(p-q)}{pq} \div c(q-p)$
 $= -\frac{1}{pq}$ ✓

$\therefore m_{OM} = -1 \times m_{MA}$

(13)

$$\text{ii) } \tan \angle MOA = m_{OM} = \frac{1}{pq}$$

$$\text{Since } \tan \angle MAX = m_{AM} = -\frac{1}{pq}$$

$$\therefore \angle MAX = \pi - \angle MOA$$

$$\therefore \angle MAO = \angle MOA$$

$$\therefore OM = MA \text{ (Equal sides of Isosceles } \triangle OMA)$$

$$\text{iii) Equation of OM is } y = \frac{1}{pq}x \text{ --- (1)}$$

To find coordinates of T:

$$xy = c^2 \text{ --- (2)}$$

$$x \cdot \frac{1}{pq}x = c^2$$

$$x^2 = c^2 pq$$

$$x = c\sqrt{pq}$$

$$y = c^2 x^{-1}$$

$$y' = \frac{-c^2}{x^2}$$

$$\text{at } x = c\sqrt{pq} \quad m_T = \frac{-c^2}{(c\sqrt{pq})^2} = \frac{-1}{pq} = m_{MA} = m_{PQ}$$

$\therefore$ Tangent at T $\parallel$ chord PQ.

(14)

Question 14

$$(a) \quad P(x) = px^3 - 3qx + r$$

$$P'(x) = 3px^2 - 3q$$

Double root is also a root of $P'(x) = 0$

$$\therefore 3px^2 - 3q = 0$$

$$x^2 = \frac{q}{p}$$

$$x = \pm \frac{\sqrt{q}}{\sqrt{p}} \text{ are possible double roots.}$$

$$P\left(\frac{\sqrt{q}}{\sqrt{p}}\right) = p \cdot \left(\frac{\sqrt{q}}{\sqrt{p}}\right)^3 - 3q \cdot \frac{\sqrt{q}}{\sqrt{p}} + r = 0$$

$$\therefore p \cdot \frac{q\sqrt{q}}{p\sqrt{p}} - 3q \cdot \frac{\sqrt{q}}{\sqrt{p}} + r = 0$$

$$\times \sqrt{p} \quad q\sqrt{q} - 3q\sqrt{q} + r\sqrt{p} = 0$$

$$\therefore r\sqrt{p} = 2q\sqrt{q}$$

squaring

$$r^2 p = 4q^3$$

Progress

(15)

$$(b) (i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\left. \begin{aligned} \text{at } P(a \cos \theta, b \sin \theta) \quad m_t &= -\frac{b^2 a \cos \theta}{a^2 b \sin \theta} \\ &= -\frac{b \cos \theta}{a \sin \theta} \end{aligned} \right\} \checkmark$$

$$(ii) m_{SM} = \frac{a \sin \theta}{b \cos \theta} \quad (MS \perp \text{tangent at } P)$$

$$\therefore y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{a \sin \theta}{b \cos \theta} (x - ae)$$

$$\therefore y = \frac{a \sin \theta}{b \cos \theta} (x - ae) \text{ is equation of MS}$$

$$\text{Equation of OP is } y = mx \text{ with } m_{OP} = \frac{b \sin \theta}{a \cos \theta}$$

$$\text{ie, } y = \frac{b \sin \theta}{a \cos \theta} x$$

Solving simultaneously,

$$\frac{b \sin \theta}{a \cos \theta} x = \frac{a \sin \theta}{b \cos \theta} (x - ae)$$

$$\frac{b^2}{a^2} x = x - ae$$

$$\therefore x \left(\frac{b^2}{a^2} - 1 \right) = -ae \quad \text{as } \frac{b^2}{a^2} = 1 - e^2$$

$$x = \frac{-ae}{1 - e^2 - 1} = \frac{-ae}{-e^2} = \frac{ae}{e^2} \quad \therefore M \text{ lies on } S\left(\frac{a}{e}, 0\right)$$

(16)

$$(c) {}^{30}C_7 \times {}^{23}C_8 \times {}^{15}C_{15} = {}^{30}C_7 \times {}^{23}C_8$$

$$\text{or } {}^{30}C_{15} \times {}^{15}C_8 \times {}^7C_7 = {}^{30}C_{15} \times {}^{15}C_8$$

or Equivalent

$$(d) i) F = -mg - \frac{v^2}{2}$$

$$\therefore m \ddot{x} = -mg - \frac{v^2}{2} \quad \text{and } m=2, g=10$$

$$\therefore 2 \ddot{x} = -20 - \frac{v^2}{2}$$

$$\begin{aligned} \ddot{x} &= -10 - \frac{v^2}{4} \\ &= \frac{-40 - v^2}{4} \end{aligned}$$

$$ii) \frac{dv}{dt} = -\frac{40 + v^2}{4}$$

$$\therefore \frac{dt}{dv} = \frac{-4}{40 + v^2}$$

$$t = -4 \cdot \frac{1}{\sqrt{40}} \tan^{-1}\left(\frac{v}{\sqrt{40}}\right) + C \quad \checkmark$$

$$\text{at } t=0, v=20 \quad \therefore 0 = -\frac{2}{\sqrt{10}} \tan^{-1}\left(\frac{20}{\sqrt{10}}\right) + C$$

$$\therefore C = \frac{2}{\sqrt{10}} \tan^{-1}\left(\frac{10}{\sqrt{10}}\right)$$

$$\therefore t = \frac{2}{\sqrt{10}} \left(\tan^{-1}\left(\frac{10}{\sqrt{10}}\right) - \tan^{-1}\left(\frac{v}{\sqrt{10}}\right) \right)$$

$$\begin{aligned} \text{at max height } v=0 \quad \therefore t &= \frac{2}{\sqrt{10}} \tan^{-1}\left(\frac{10}{\sqrt{10}}\right) \quad \checkmark \\ &= 0.799 \dots \\ &\approx 0.8 \text{ seconds} \end{aligned}$$

(17)

$$\text{iii)} \quad v \cdot \frac{dv}{dx} = -\frac{40 - v^2}{4}$$

$$\therefore \frac{dx}{dv} = \frac{-4v}{40 + v^2}$$

$$x = -2 \int \frac{2v}{40 + v^2} dv$$

$$= -2 \ln(40 + v^2) + C$$

$$\text{at } v=0, x=20\text{m} \quad \therefore C = 2 \ln 40$$

$$\therefore x = 2 \ln \left(\frac{40 + v^2}{40} \right) \quad \checkmark$$

$$\text{at } v=0, x=H \text{ (max height)}$$

$$\therefore H = 2 \ln 11 \quad \checkmark$$

iv) let speed of projection be u

$$\therefore x = 2 \ln \left(\frac{40 + u^2}{40} \right) \text{ from part iii)} \quad \checkmark$$

$$\text{at } v=0, x=2H = 4 \ln 11$$

$$\therefore 4 \ln 11 = 2 \ln \left(\frac{40 + u^2}{40} \right)$$

$$\therefore 121 = \frac{40 + u^2}{40}$$

$$u^2 = 4900$$

$$u = 40\sqrt{3} \text{ m/s} \quad u > 0 \quad \checkmark$$

$$\approx 69.282 \dots \text{ m/s}$$

(18)

Question 15

$$\text{a) i) let } f(x) = x - \ln(1+x) \\ f'(x) = 1 - \frac{1}{1+x} > 0 \text{ for all } x > 0 \quad \checkmark \\ \text{as } 1+x > 1$$

$$\text{and } f(0) = 0$$

$$\therefore f(x) > 0 \text{ as } f(x) \text{ is monotonic increasing for all } x > 0 \quad \checkmark \\ \text{and } f(0) = 0$$

$$\therefore x - \ln(1+x) > 0$$

$$\therefore x > \ln(1+x)$$

ii) When $n=1$

$$\text{L.H.S} = \frac{1}{1} \quad \text{R.H.S} = \ln(1+1) \\ = \ln 2$$

$$= 1 \quad \approx 0.69 \dots \quad \checkmark$$

$\therefore$ statement is true when $n=1$

Assume statement true for $n=k$

$$\text{i.e. } \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} > \ln(1+k)$$

Let $n=k+1$

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} + \frac{1}{k+1} - \ln(1+k+1)$$

$$> \ln(1+k) - \ln(1+k+1) + \frac{1}{k+1} \quad \text{by assumption} \quad \checkmark$$

$$> \ln(1+k) - \ln(1+k+1) + \ln\left(1 + \frac{1}{k+1}\right) \quad \text{by part i)}$$

$$= \ln\left(\frac{1+k}{1+k}\right) + \ln\left(\frac{k+2}{k+1}\right) \quad \downarrow \text{progress} \quad \checkmark$$

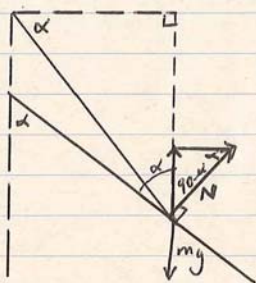
$$= \ln\left(\frac{1+k}{1+k} \cdot \frac{k+2}{k+1}\right)$$

$$= \ln 1 = 0$$

(19)

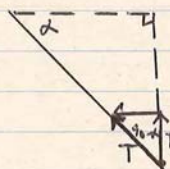
Question 15 Continued

i)



From diagram: Vertical component of N is
 $N \cos(90-\alpha) = N \sin \alpha$.

ii)



Vertical component of T is $T \sin \alpha$ ✓

Resolving forces vertically:

$$T \sin \alpha + N \sin \alpha = mg$$

$$\therefore T + N = \frac{mg}{\sin \alpha} \quad \text{--- (1)}$$

Sum of radial forces = $m\omega^2 r$

$$\therefore T \cos \alpha - N \cos \alpha = m\omega^2 r \quad \text{but } \cos \alpha = \frac{r}{L}$$

$$\therefore T - N = m\omega^2 L \quad \text{--- (2) ✓}$$

iii) ① - ②:

$$2N = \frac{mg}{\sin \alpha} - m\omega^2 L \quad \checkmark$$

at $N = 0$

$$0 = \frac{g}{\sin \alpha} - \omega^2 L$$

$$\therefore \omega = \sqrt{\frac{g}{L \sin \alpha}} \quad \checkmark \quad \omega > 0$$

(20)

Question 15

$$(c) \quad i) \quad \left. \begin{aligned} a^2 + b^2 &\geq 2ab \\ a^2 + c^2 &\geq 2ac \\ a^2 + d^2 &\geq 2ad \\ b^2 + c^2 &\geq 2bc \\ b^2 + d^2 &\geq 2bd \\ c^2 + d^2 &\geq 2cd \end{aligned} \right\} \quad \checkmark$$

Adding $3a^2 + 3b^2 + 3c^2 + 3d^2 \geq 2ab + 2ac + 2ad + 2bc + 2bd + 2cd$
 $3(a^2 + b^2 + c^2 + d^2) \geq 2(ab + ac + ad + bc + bd + cd)$ ✓

ii) Now $\sum a^2 = (\sum a)^2 - 2(\sum ab)$

So $3[(a+b+c+d)^2 - 2(ab+ac+ad+bc+bd+cd)] \geq 2(ab+ac+ad+bc+bd+cd)$ ✓

$\therefore 3(1)^2 \geq 8(ab+ac+ad+bc+bd+cd)$ as $a+b+c+d=1$

$\therefore ab+ac+ad+bc+bd+cd \leq \frac{3}{8}$

(21)

Question 16

$$(a) \quad P(7) = \frac{{}^1C_1 \times {}^{39}C_5}{{}^{40}C_6} = \frac{3}{20}$$

$$\therefore P(\bar{7}) = \frac{17}{20}$$

$$i) \quad P(X=2) = {}^5C_2 \left(\frac{3}{20}\right)^2 \left(\frac{17}{20}\right)^3$$

$$= 0.1382 \quad (4 \text{ d.p.}) \quad \checkmark$$

$$ii) \quad P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - [P(X=0) + P(X=1)]$$

$$= 1 - \left[{}^5C_0 \left(\frac{3}{20}\right)^0 \left(\frac{17}{20}\right)^5 + {}^5C_1 \left(\frac{3}{20}\right)^1 \left(\frac{17}{20}\right)^4 \right]$$

$$= 0.1648 \quad (4 \text{ d.p.}) \quad \checkmark$$

iii)

$$P(7 | \text{five of the numbers 1 to 6}) = \frac{{}^1C_1 \times {}^6C_5}{{}^{40}C_6} = \frac{1}{639730}$$

$$P(E) = 1 - P(7 \text{ being the highest number in none of the 5 games})$$

$$= 1 - {}^5C_0 \left(\frac{1}{639730}\right)^0 \left(\frac{639729}{639730}\right)^5$$

$$= 7.82 \times 10^{-6} \quad (3 \text{ sig. fig.}) \quad \checkmark$$

(22)

Question 16

- (b) (i) If $\angle CDB = \alpha$
 then $\angle DAC = \alpha$ (Angle in the alternate segment theorem)
 and $\therefore \angle DBC = \alpha$ (Angles on the circumference standing on the same arc are equal)
 $\therefore \triangle BCD$ is isosceles (Base $\angle$'s equal).

- (ii) $CD^2 = AC \cdot CK$ (Square on tangent).
 but $CD = BC$ (Equal sides of isosceles $\triangle BCD$)
 $\therefore BC^2 = AC \cdot CK$
 $\therefore BC$ must be a tangent to circle AKB .

- (c) (i) Consider first

$1 - x^2 + x^4 - x^6 + \dots + x^{4n}$ which is a G.P. with $a=1$, $r=-x^2$ and $2n+1$ terms $\checkmark$

$$S = \frac{a(1-r^n)}{1-r}$$

$$= \frac{1[1 - (-x^2)^{2n+1}]}{1+x^2}$$

$$= \frac{1+x^{4n+2}}{1+x^2} > \frac{1}{1+x^2} \quad \text{as } 1+x^{4n+2} > 1 \text{ for } x > 0.$$

$$\therefore \frac{1}{1+x^2} < 1 - x^2 + x^4 - x^6 + \dots + x^{4n}$$

Consider $1 - x^2 + x^4 - x^6 + \dots + x^{4n} - x^{4n+2}$ $a=1$, $r=-x^2$ with $2n+2$ terms $\checkmark$

$$= \frac{a(1-r^n)}{1-r}$$

$$= \frac{1[1 - (-x^2)^{2n+2}]}{1+x^2}$$

$$= \frac{1 - x^{4n+4}}{1+x^2} < \frac{1}{1+x^2} \quad \text{as } 1 - x^{4n+4} < 1 \text{ for } x > 0$$

(23)

$$\text{So } 1 - t^2 + t^4 - t^6 + \dots + t^{4n} - t^{4n+2} < \frac{1}{1+t^2}$$

$$\therefore S < \frac{1}{1+t^2} + t^{4n+2}$$

$$\therefore \frac{1}{1+t^2} < 1 - t^2 + t^4 - t^6 + \dots + t^{4n} < \frac{1}{1+t^2} + t^{4n+2}$$

$$\text{ii) } \int_0^x \frac{1}{1+t^2} dt < \int_0^x 1 - t^2 + t^4 - t^6 + \dots + t^{4n} dt < \int_0^x \frac{1}{1+t^2} + t^{4n+2} dt$$

$$\left[\tan^{-1} t \right]_0^x < \left[t - \frac{t^3}{3} + \frac{t^5}{5} - \frac{t^6}{6} + \dots + \frac{t^{4n+1}}{4n+1} \right]_0^x < \left[\tan^{-1} t + \frac{t^{4n+3}}{4n+3} \right]_0^x$$

$$\therefore \tan^{-1} x - \tan^{-1} 0 < \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^6}{6} + \dots + \frac{x^{4n+1}}{4n+1} \right) - (0) \quad \text{✓ progress}$$

$$< \left(\tan^{-1} x + \frac{x^{4n+3}}{4n+3} \right) - 0$$

$$\therefore \tan^{-1} x < x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^6}{6} + \dots + \frac{x^{4n+1}}{4n+1} < \tan^{-1} x + \frac{x^{4n+3}}{4n+3}$$

$$\text{iii) as } n \rightarrow \infty, \frac{x^{4n+3}}{4n+3} \rightarrow 0 \text{ as } 0 \leq x \leq 1.$$

$$\therefore x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \text{ has an upper}$$

and lower bound of $\tan^{-1} x$. ✓

$$\therefore \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$\text{iv) let } x=1 \therefore \tan^{-1}(1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \quad \text{✓}$$

$$\therefore \pi = 4 \times \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$